

RealitySketch: Augmented Reality Sketching for Real-time Embedded and Responsive Visualizations

Ryo Suzuki
University of Calgary
Calgary, AB, Canada
ryo.suzuki@ucalgary.ca

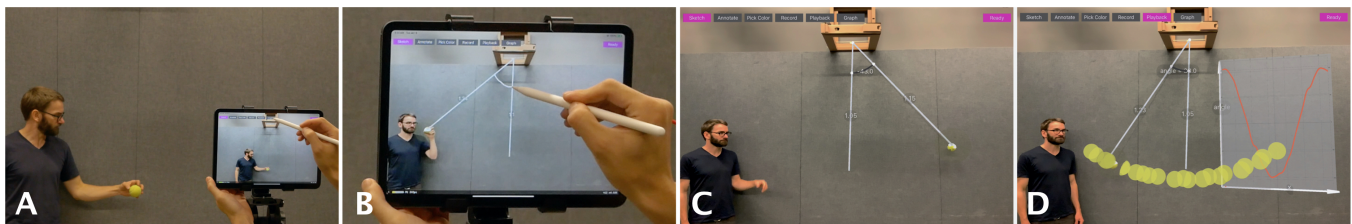
Rubaiat Habib Kazi
Adobe Research
Seattle, WA, USA
rhabib@adobe.com

Li-Yi Wei
Adobe Research
San Jose, CA, USA
lwei@adobe.com

Stephen DiVerdi
Adobe Research
San Jose, CA, USA
diverdi@adobe.com

Wilmot Li
Adobe Research
Seattle, WA, USA
wilmotli@adobe.com

Daniel Leithinger
University of Colorado Boulder
Boulder, CO, USA
daniel.leithinger@colorado.edu



RealitySketch enables the user to draw and visualize physical phenomena like a pendulum's motion through real-time sketching: A) Select the ball to track, then draw a line and bind it with the physical ball. B) Draw a vertical line and an arc to parameterize the angle between the tracked ball and the center line. C) The sketched elements dynamically respond when the pendulum swings. D) Responsive graphics record and visualize the motion of the pendulum.

ABSTRACT

In this Real-Time Live, we demonstrate RealitySketch, an augmented reality interface for sketching interactive graphics and visualizations [Suzuki et al. 2020]. In recent years, an increasing number of AR sketching tools enable users to draw and embed sketches in the real world. However, with the current tools, sketched contents are inherently *static*, floating in mid air without responding to the real world. This paper introduces a new way to embed *dynamic* and *responsive* graphics in the real world. In RealitySketch, the user draws graphical elements on a mobile AR screen and binds them with physical objects in real-time and improvisational ways, so that the sketched elements dynamically move with the corresponding physical motion. The user can also quickly visualize and analyze real-world phenomena through responsive graph plots or interactive visualizations. This paper contributes to a set of interaction techniques that enable capturing, parameterizing, and visualizing real-world motion without pre-defined programs and configurations. Finally, we demonstrate our tool with several application scenarios, including physics education, sports training, and in-situ tangible interfaces.

CCS CONCEPTS

• **Human-centered computing** → **Human computer interaction (HCI)**.

KEYWORDS

augmented reality; embedded data visualization; real-time authoring; sketching interfaces; tangible interaction;

ACM Reference Format:

Ryo Suzuki, Rubaiat Habib Kazi, Li-Yi Wei, Stephen DiVerdi, Wilmot Li, and Daniel Leithinger. 2021. RealitySketch: Augmented Reality Sketching for Real-time Embedded and Responsive Visualizations. In *SIGGRAPH Asia 2021 Real-Time Live! (SA '21 Real-Time Live!)*, December 14–17, 2021, Tokyo, Japan. ACM, New York, NY, USA, 1 page. <https://doi.org/10.1145/3478511.3491313>



Figure 1: In-situ creation of a tangible UI slider.

REFERENCES

Ryo Suzuki, Rubaiat Habib Kazi, Li-Yi Wei, Stephen DiVerdi, Wilmot Li, and Daniel Leithinger. 2020. RealitySketch: Embedding Responsive Graphics and Visualizations in AR through Dynamic Sketching. In *Proceedings of the 33rd Annual Symposium on User Interface Software and Technology*. ACM.

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

SA '21 Real-Time Live!, December 14–17, 2021, Tokyo, Japan

© 2021 Copyright held by the owner/author(s).

ACM ISBN 978-1-4503-9072-9/21/12.

<https://doi.org/10.1145/3478511.3491313>